

Task16

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In this spike we had to do a layered FSM with a soldier Agent. The FSM manages high- level modes and low levels.

High level states handle patrolling and attacking when enemies are spotted within a certain range.

Low level states handle seeking waypoint, arriving to the waypoint, and then seeking again.

As well as shooting, being out of ammo, reloading, and shooting again.

Patrol triggers enemy within radius, which makes it Attack.

Attack triggers dead enemy, which makes it Patrol.

Seek waypoint triggers reaching the waypoint, which makes it Arrive at waypoint

Arrive triggers velocity near zero, which makes it seek the next waypoint

Shoot triggers ammo = 0, which makes it reload.

Finally, Reload triggers the timer complete, which allows it to Shoot again.

To actually implement it, the FSM has 2 classes, FSMState which has a dictionary of transitions, and FSM collects the states and tracks the current one. All the states are built within Soldier._build_fsm()

After each frame, Soldier.update checks FIRST for any possible high level state, if they are not currently available, then it checks for the low level states.

The visuals have the stationary enemies that are spawned by pressing E, the soldier stands still when shooting, the enemies use a simple health model of 30 HP (and each hits does 10 damage).

My principal challenge was getting the FSM transitions to actually fire. I would get my other states to “work” but I couldn’t test properly because the soldier would switch to Attack mode, but then it would never Shoot. In the end, I didn’t properly write calculate() for the specific attack states.